Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Vectors

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the vector $\overrightarrow{AB}$ given $A(1, 2, 3)$ and $B(4, 0, 1)$. **[2]**

2. Find the magnitude of $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$. **[2]**

3. Find a unit vector in the direction of $3\mathbf{i} + 4\mathbf{j}$. **[3]**

4. Find the scalar product of $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. **[3]**

5. Find the angle between $\mathbf{a} = \mathbf{i} + \mathbf{j}$ and $\mathbf{b} = \mathbf{i} - \mathbf{j}$. **[4]**

6. Find the vector equation of the line through $A(1, 0, 2)$ parallel to $\mathbf{i} + \mathbf{j} - \mathbf{k}$. **[3]**

7. Find the vector equation of the line through $A(1, 2, 3)$ and $B(4, 5, 6)$. **[3]**

8. Determine whether the lines $\mathbf{r} = \mathbf{i} + t(\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + s(\mathbf{i} + \mathbf{k})$ intersect. **[5]**

9. Find the Cartesian equation of the plane through $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$. **[4]**

10. Find a normal vector to the plane $2x - y + 3z = 5$. **[1]**

11. Find the distance from the point $(1, 1, 1)$ to the plane $x + y + z = 0$. **[4]**

12. Show that $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j}$ are perpendicular. **[2]**

13. Find the vector product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{j} + \mathbf{k}$. **[4]**

14. Find the area of the triangle with vertices O, $\mathbf{a}$, $\mathbf{b}$ where $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{k}$. **[4]**

15. Write the Cartesian equation of the line $\mathbf{r} = (1 + t)\mathbf{i} + (2 - t)\mathbf{j} + 3\mathbf{k}$. **[3]**

Section B – Hard (Q16–25)

16. Find the shortest distance between the skew lines $\mathbf{r} = \mathbf{i} + t(\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + s(\mathbf{i} - \mathbf{k})$. **[6]**

17. Find the equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, $C(0, 0, 1)$. **[4]**

18. Find the angle between the planes $x + y + z = 1$ and $x - y + z = 2$. **[4]**

19. Find the foot of the perpendicular from $(1, 2, 3)$ to the plane $x + y + z = 6$. **[5]**

20. Determine whether the lines $\mathbf{r} = \mathbf{a} + t\mathbf{d}_1$ and $\mathbf{r} = \mathbf{b} + s\mathbf{d}_2$ are parallel, intersecting or skew given $\mathbf{d}_1 \cdot \mathbf{d}_2 = 0$ and $(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2) \neq 0$. **[3]**

21. Find the volume of the parallelepiped formed by $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = \mathbf{j} + \mathbf{k}$, $\mathbf{c} = \mathbf{k} + \mathbf{i}$. **[4]**

22. Find the equation of the plane through the line $\mathbf{r} = \mathbf{i} + t(\mathbf{j} - \mathbf{k})$ and the point $(1, 1, 1)$. **[5]**

23. Find the reflection of the point $(1, 0, 0)$ in the plane $x + y + z = 1$. **[5]**

24. Parametric equations of a line are $x = 1 + t$, $y = 2 - t$, $z = 3 + 2t$. Find the point where it meets the plane $x + y + z = 10$. **[4]**

25. Prove that the shortest distance between two skew lines is $\dfrac{|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$. **[5]**

End of Worksheet • *Show all working clearly.*